Chapter 2 — Surveys, observation, and scraping

Knowing a source type is not enough, you still need an instrument

Learning objectives

- Match surveys, observation
- Scraping to typical research questions, name the main watch-outs for each method
- Explain why instrument and sample planning precede platform choice

Choosing a collection method

- Once source type is chosen, teams select concrete methods
- Surveys and questionnaires gather structured responses from people
- Observation captures behavior in context
- Web scraping extracts content from pages that lack APIs
- Mixed designs combine instruments when no single method answers both the what and the why
- This part focuses on when each method fits; APIs are covered in the next part

Surveys and questionnaires

- Public-opinion work
- Modes include online forms, face-to-face interviews, telephone contact
- Strengths include large sample reach, standardized wording, and quantifiable outputs
- Weaknesses include socially desirable answering, low response rates without incentives

Example 2.5 — Campus Wi-Fi satisfaction survey plan

- Example 2.5 — hands-on module
- Example 2.5 shows a short instrument plan before any survey platform is chosen
- An IT office wants to know which buildings have unreliable Wi-Fi
- Device type, plus an open comment box for locations the closed items miss
- Explore the chapter example module
- View files: `modules/chapter2/example5/`

Observation

- Observation gathers data by watching behavior or events in their natural setting
- Participant observation embeds the researcher in the activity
- Often hard to generalize from small settings
- Teams frequently combine observation with surveys so depth and breadth reinforce each

Web scraping

- Web scraping uses programs to extract text, tables, or media from HTML pages
- It is useful when a site exposes public content but no API
- May trigger anti-bot defenses
- Responsible projects document robots policies, throttle requests

Example 2.6 — Choosing scraping for public course catalogs

- Example 2.6 — hands-on module
- Example 2.6 stays at the decision level
- A research team needs weekly snapshots of publicly listed university course titles and
- They scrape the HTML timetable, store course code and title as structured rows
- Explore the chapter example module
- View files: `modules/chapter2/example6/`

Takeaways

- Surveys fit attitudes and self-report at scale
- Each method carries distinct bias and policy risks
- Design the instrument and sampling plan first

Next

- Complete the quiz for this part
- Complete the quiz